Transforming Game-based Learning into Real World Executive Functioning Skills

- Presented at the 2017 Executive Function and Learning West Coast Conference
- Randy Kulman, Ph.D.
- randy@learningworksforkids.com
- Twitter @rkulman, @lw4k
- Website learningworksforkids.com
- FB- LearningWorks for Kids
- Pinterest lw4k

Takeaways

- The importance of EFs for kids
- Traditional strategies to improve EFs
- Play and Games as tools to improve EFs
- The research on game-based learning (GBL) of EFs
- Generalization- the holy grail of GBL
- Identify popular games/apps to improve EFs
- Making game-based EFs into real world skills

What are Executive Functions?

- The cognitive command and control systems of the brain.
- The CEO of the brain
- The manager, organizer, and coordinator of internal and external information.
- A collection of brain processes that guide thought and behavior.
- Capacities that enable independent, purposeful, goal-directed behavior.
- Critical-thinking and self-regulatory skills

News Flash: Psychologists do NOT Agree on a Definition of Executive Functions

- Maintain a problem-solving set for future goals - Welsh and Pennington
- Processes responsible for purposeful, goal-directed behavior - Gioia, Isquith, Guy
- Cognitive and behavioral skills responsible for highest levels of human functioning – Lezak
- Cognitive control processes that regulate and shape behavior - Friedman
- Self-regulatory actions including self-management of time, self-organization/problem solving, self-discipline/inhibition, self-motivation, self-activation/concentration - Russell Barkley

Executive Functions - Conductor of the Orchestra

- Conductors:
- Connect to all aspects of the music
- Integrate all the individual musicians and their instruments.
- Organize, plan, and direct the skills of musicians and their instruments
- Use memory of previous experiences, identify strengths and weaknesses, and make adjustments to volume and emphasis

Executive Functions at LW4K

- Task initiation
- Sustained attention
- Task persistence
- Response inhibition
- Affective regulation
- Time management
- Working Memory
- Organization
- Planning
- Flexibility
- Social Thinking
- Metacognition
- (Based upon Dawson and Guare)

Neuroscience of Executive Functions

- Executive functions are not located simply in the prefrontal cortex.
- Executive functions involve neural networks tasked with coordinating and processing internal and external information.
- The prefrontal cortex sends reciprocal connections to many other areas of the brain involving executive functions.
- It is best to view executive functions as an interconnected coordination process across wide ranges of areas of the central nervous system (including but not restricted to the parietal, temporal, and occipital cortex, basal ganglia, cerebellum, amygdala, and hippocampus).
- http://learningworksforkids.com/executive-functions/

Problems with Working Memory

Executive Functioning Skills in School

- Doing homework – organization, planning, focus, time management
- Phonological awareness – working memory
- Written language - organization, focus, planning
- Math word problem - working memory, focus, planning

The Importance of Teaching Executive Functioning Skills

- Preschoolers who learn EF's perform better academically (Diamond, McClelland)
- Improving working memory enhances reading, reduces ADHD symptoms (Cogmed research)
- Self-control enhances future life success (Marshmallow test)
- Protects children from mental health issues
- Problem solving, personal growth, relationships, and education
- A similar concept to other ideas such as grit, character development, social/ emotional learning, emotional intelligence that are viewed as core ingredients for healthy adaptation
- Major role in decision making, problem solving, parenting skills, and overall happiness

Organizational Skills Needed

Traditional Interventions for Improving Executive Functions

- Many kids need to be taught skills, rather than picking them up on their own
- May require teaching or modeling of the skills by teachers, mentors and parents, e.g., organizing HW or backpack
- Sustained practice - requires salient rewards, reminders, recognition, and accomplishment, kids for whom the morning routine is like the first time they did it
- Understanding and ownership of the goal of developing a new skill on the part of the child
- Understanding how this new skill will help the child
- Teaches a child to use executive skills and apply them in new situations
- Generalization strategies to help apply it to new situations

Accommodations to Improve Executive Functions

- Definition - modifications that promote learning and success
- Not too different from the kid who needs glasses or can't handwrite
- Kids who work slowly, are forgetful, need to move, have sensory issues
- We assume that if it is a clear physical disability, e.g., hearing or visual issues, we should adjust and accommodate, but that kids who are inherently disorganized, distracted, or anxious are not trying hard enough
- Motivation vs. ability? Not always quite so simple, e.g., ADHD kids and their interests.
- Classroom accommodations - how it is taught, how it is assigned, how it is evaluated, and the pace of teaching

Cognitive Engagement and Tech

Video Games and Learning

- Play = Learning: Psychological perspectives from Piaget, Elkind, Brown, Gray, Singers
- Play- Test and explore observations, learn about relationships, cooperation, frustrations, success (SEL)
- Play- How kids learn about expressing selves, restraining impulses, taking perspectives, imitating, and creating
- Play- learn to plan, think of the future, executive functions
- Digital play = learning: Gee, Squire, organizations such as Serious Play, Games for Change and Health, University of Wisconsin

Why use Video Games, Apps, and Digital Technologies to Improve Efs

- Engaged kids
- Catch them with what they are already doing
- Willing practice
- Sustained attention and effort
- Uses skills (EFs) that are crucial to 21st century success
- Research support for games and learning

Motivation, Metacognition and Game-based Learning

- Games are social, and learning occurs with others
- Games foster communication amongst kids
- Games get special needs kids to participate
- Games get kids to think about what they are doing
- Flow and engagement
- Attention, practice, and effort lead to learning

Executive Functions in Game Play

- Planning- anticipating an enemy attack in a FPS
- Working Memory- Recall a series of moves in battle
- Flexibility- Change approach with new levels
- Self- Awareness- learning from mistakes
- Self-Control- controlling frustration to progress
- Organization- manage inventories and supplies

Specific Games and Learning

- Academics - Reading, math fluency, topical areas
- Cognitive skills - Processing speed, attention abilities
- Job-related skills - e.g., Laparoscopic surgery, military functions, leadership
- Social-emotional functioning - Social skills
- Executive functions - Working memory, cognitive flexibility, focusing

Social Thinking Skills and Tech

Research on executive functions and positive effecs of video games

- Increasing Proccessing Speed (Green and Bavelier 2009)
- Improving Working Memory (Klinberg et al. 2007)
- Increasing Pro-Social Behavior in Children (Gentile et al 2009)
- Improving Social Involvment (Ferguson, 2010)
- Building Brain Regions (Kuhn and colleagues study of Tetris)

Hyperlinks Continued…

- *Starcraft*: Improves Brain Flexability (Glass et al. 2013)
- *Rayman Raving Rabbids:* Improves Reading Fluency (Franceschini, Gori, 2013)

Research on Executive Functioning Skills and Video Games

- LW4K pilot research on differentiated instruction, targeting areas of EF weakness with video games
- Combination of board and video games improve fluid reasoning and processing speed (Mackey, 2011)
- Working memory video games improve WM, fluid reasoning skills (Cogmed) Intensity/duration
- Computer-based training improves executive attention in preschoolers (Rueda, 2005)
- Video game-like math and reading programs improve learning, reduces attention symptoms
- Games (non-video) increase cognitive load - teaching tool by parents reduces ADHD signs in preschoolers (Halperin, 2012)
- Study on games to improve working memory (Davis, 2011)

Research – Negative Impact of Video Games on Children

- Increasing levels of obesity with screen-based time, primarily television
- Poor psychological adjustment in kids who play more than 3 hours per day - Pediatrics report
- Violence and video games - Bushman data
- 7 hours 38 minutes per day (2001) of digital-media time - ignoring other activities - Kaiser Foundation data- 9.5 hrs teens Commonsense media study
- Video game addiction 3- 8% cited-DSM-V category internet gaming disorder

How Popular Games and Apps Improve Executive Functions

- **Practice** – primarily with games with the skill is used repeatedly in order to achieve goals
- **Support**, primarily apps where a skill is scaffold by the functionality
- **Mastery**, primarily apps and games with tice

Minecraft

- EF and SEL skills used
- Flexibility
- Organization
- Time management
- Planning/decision maki
- Connect/relationship

Minecraft

PlayBook: Minecraft

GAME FACTS | HOW IT HELPS | PLAY TOGETHER | MAKE IT WORK | MATHEMATICS | AUTISM | ADHD

Title: Minecraft
Game Type: Action
ESRB Rating: Everyone
Platform/Console: Android, iPad, iPhone, Mac, Online, PC, Xbox 360
LWK Recommended Age: 6+
Thinking Skills Used: Flexibility, Focus, Organization, Planning, Time Management
Academic Skills Used: Mathematics
Buy Now/Play Now: iTunes / Minecraft.net / Google Play

LQ: 9.5
Brain grade: 9.4
Fun score: 9.6
What is this?

Favorite

Game Description

Minecraft is an open-world game unlike any other. Players are placed in a borderless, randomly generated land with no supplies, directions, or objective. It is up to them to decide what to do and how to do it. Players collect materials from the world around them in order to 'craft' items and build whatever their mind can imagine. The concept is similar to playing with LEGOs, only the amount of blocks is limitless. There are forests to explore, mountains to climb, oceans to cross, and dungeons to discover. When the sun goes down, the monsters appear, and players must fend them off or build shelter for protection. Objectives are imagined by players, allowing them to truly create their own virtual world, mining materials and building tools to better help them do so. Due to some minimal violence and the initial complexity of the game, *Minecraft* is recommended for ages 6 and up.

Continue

Angry Birds Go

- EF and SEL skills used
- Flexibility
- Focus
- Time management
- Working memory
- Manage emotions

But Games are Not Enough!!!

- Studies question the far transfer (generalization) of VG brain training programs to real world
- The key to success is effective mediation (can be done in the game) teaching from the game
- Teachers (including peers, parents, and embedded instruction) make the connection between game-based learning and real-world skills
- Actual learning requires knowledge of the skill, an understanding of how and when to use it,

How Well do Game-based Skills Transfer to the Real World?

- It is all about generalization!
- Games alone results in modest improvements in real-world executive skills
- Children with learning and attention problems have problems in generalizing strategies
- Games as a teaching tool with mediators

The Importance of Generalization

- *A definition* - The ability to take something you've learned in one place and apply it in another place.
- *Why it is so important?* - It is a key to making classroom (or home-based) learning into real-world learning.
- *Where does it help children?* - Everywhere!
- Generalization insures that individuals can utilize the skills they have learned in one environment in various settings, with other people, and with different materials

Motivated and Attentive to Tech

How to make GBL into RWL

- The Sharp Brains model for brain training
  1. Train a specific brain-based skill
  2. Target area of weakness with specific tools/ tech
  3. Make games/ tools adaptive
  4. Insure intensity and duration
  5. Have long-term maintenance training
  6. Include generalization training

The LW4K Model

- Integrate strategic teaching principles
- Explicit goals, partnerships with child, previewing, engagement, individualization, teachable moments
- Detect, Reflect, Connect

Effective Game-based Learning

- Embedded teaching in games
- Practice in games for real-world skills
- Going outside of the game as a part of the game
- Collaborative game play - learning social skills and leadership
- Discussion and modding outside of the game

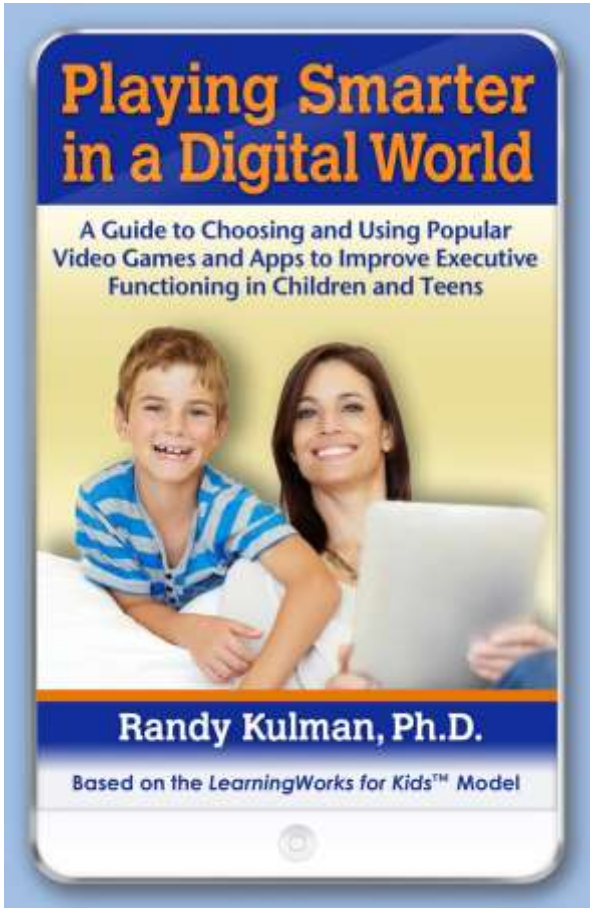
Improving Generalization by Building it into the Games

- New games such as **IF** (If You Can) for improving social awareness and self-control skills
- Innovations in technology such as a Akilli's Project Evo and Mighteor from Neuromotion
- Cogmed Working Memory training
- Lumosity/ other brain-training tools
- Still very limited, and the critics of brain training tools are quite vocal
- Using VR to make generalization easier
- Guided game play- LW4K Minecraft server

Limitations of Using Popular Games and Technologies to Improve Executive Functions

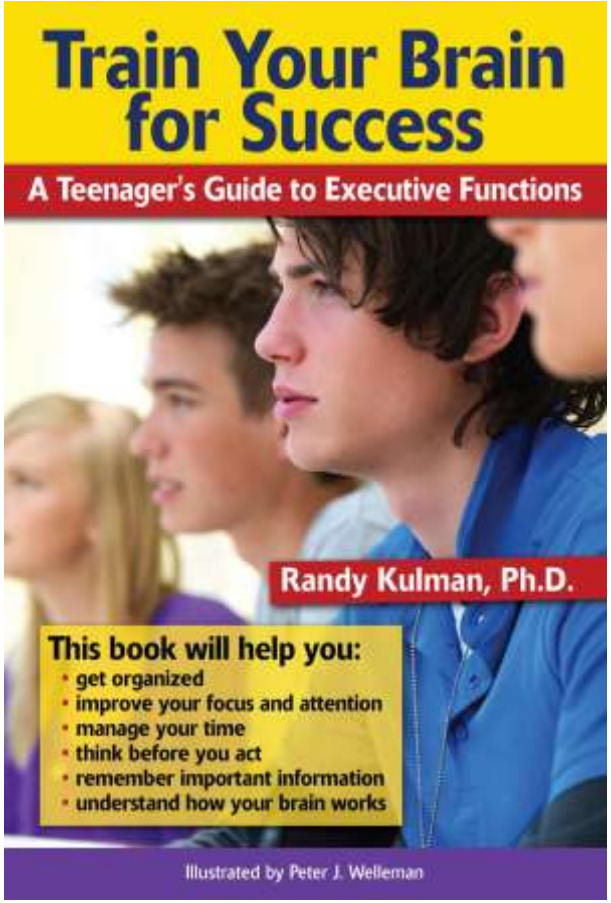
- Lack of peer-reviewed research
- Difficulty in tracking engagement and progress in the game and relating it to real-world skills and improvement
- Skills that are practiced may be less focused or lack intensity and duration
- Extraneous information and objectives that occur within the game
- Focuses on fun and not skill development
- Attitude of educators and researcher is often negative and skeptical
- No evidence of generalizability

Playing Smarter in a Digital World: A guide to Choosing and Using Popular Video Games and Apps to Improve Executive Functioning in Children and Teens



"The book's clearly articulated strategies help parents use digital media in a more effective manner and, at the same time, set effective limits and implement a healthy "play diet" for their children. A section devoted to exploring specific strategies for using digital media with children in specific populations—such as children affected by ADHD, autism spectrum and learning disorders, and other mental health and educational issues—is also featured, as is a list of specific games, apps, and tools to make game-based learning most effective."

Train Your Brain for Success: A Teenager's Guide to Executive Functions



- "Executive functions are a set of thinking, problem-solving, and self-control skills that tell the brain what to do, and this book demonstrates the ways kids use executive functions in school, at home, and in their other activities and shows how these skills can be improved through sustained effort. "

Thank You

- **Contact info:**
- Learningworksforkids.com
- www.facebook.com/LearningWorksForKids
- Twitter- @lw4k
- @rkulman
- Pinterest.com/lw4k
- 401-515-2006
- randy@learningworksforkids.com